TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study No. 4

Team Details

Team Innovex

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THINK. BUILD. CREATE. WIN

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ClauseLink: every label element traced to the clause that requires it

The problem

A single product change may require updates to **30+ labels** and **20+ languages** (p.4).

Causes of human error named in the case study (p.4):

- Manual data entry and copying between ERP, PLM, labelling software and regulatory databases
- Requirements that keep changing: UDI updates, new symbols, MDR changes
- Reviewers comparing old versions, new versions and regulatory documents by hand
- Translation inconsistencies; visual inspection that depends on people

FDA device recall records with a labeling root cause

5.59%	of 59,481 records, all years
3.95%	of 10,299 records, 2021 to 2025
135 of 407	2021 to 2025 labeling records cite "Labeling Change Control"

Source: openFDA device recall API, five labeling root-cause categories, data as of 2026-10-15. Records are per product; openFDA says to treat them as unvalidated.

Architecture: the graph is the system of record

Inputs: PLM and ERP data, regulatory sources, existing approved labels

M2 Regulatory intelligence
Cited rule changes; RA approves them.

M0 Knowledge graph
- **Requirement** — the clause that requires it
- **Label element** — LOT, UDI, warning, symbol
- **Source data** — PLM or ERP attribute
- **Artwork** — market, language, level
- **Approval** — change order, e-signature

M1 Change-impact engine
One query to every affected label.

M5 Authoring and translation
Approved phrases only; new text is a draft.

M6 UDI-DI change detector
MDR Annex VI 3.9 triggers; US cases go to RA.

M3 Compliance checker
Rules decide; LLM notes are flagged.

M4 CV verification
Printed UDI vs DataMatrix, symbols.

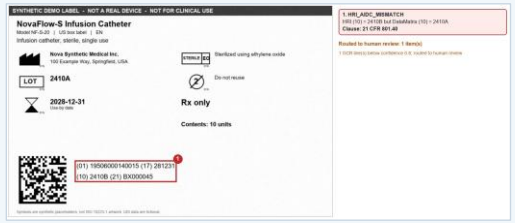
Human e-signature gate
RA and QA sign; AI never approves.

Release: signed version, change order and audit trail written back to the graph

Coverage of the 10 pain points (p.5)

Pain point	M0	M1	M2	M3	M4	M5	M6	WF
1. Time-Consuming Manual Process	○	●		●	●	○		○
2. Frequent Regulatory Updates	○	●	●					
3. Symbol Management	●	○	○	●	●			
4. Multiple Sources of Truth	●							
5. Expertise Dependency	●		○	●		○	○	
6. Manual Redlining		○			●	●		
7. Repetitive Review				●	●			○
8. Traceability	●	○						●
9. Variant Explosion	●	●				●		
10. Human Error	○			●	●	●	○	

● primary ○ supporting. M0 graph, M1 to M6 modules, WF workflow with e-signature and audit trail. Every pain point has a primary module.



Prototype check

Synthetic test: 4/4 seeded defects caught, 0 false flags across 10 proofs; uncertain reads go to a human.

Shown: printed lot differs from the DataMatrix.

Process coverage

18-step flow plus a change trigger: 8 automated, 7 AI-assisted, 4 human (stakeholder review and 3 approvals).

Estimate: 42.9% to 88.1% less manual effort per change (48.5 to 225.5 h saved) for the 30-label, 20-language example. Assumptions in Annexure B.

Guardrails

- **AI never approves.** RA and QA e-sign every release.
- **Part 11 audit trail:** time-stamped record of every change (21 CFR 11.10(e)).
- **Risk-based review:** full review for UDI, warnings and sterile data.
- **Validated** as quality-system software (QMSR / ISO 13485, FDA CSA 2026).

Annexure A: Supporting detail, process map today and with ClauseLink

#	Today (case study, p.3)	With ClauseLink	Mode	Module
0	***(new) Change trigger***	PLM/ERP change, approved rule change or new market opens a change record	**Automated**	M0, M2
1	**Product / Design Information**	Connectors sync product attributes with source ID and version	**Automated**	M0
2	**Product Understanding**	Attributes that drive requirements shown with their source; extracted values need owner confirmation	**AI-assisted**	M0
3	**Identify Applicable Markets**	Read from registration and ERP data; RA confirms any new market	**AI-assisted**	M0
4	**Regulatory Requirement Assessment**	Graph query returns the applicable clauses; new rules enter only after RA approval	**AI-assisted**	M0, M2
5	**Gap Assessment**	Rule diff of required vs present elements, per label, market and level	**Automated**	M3
6	**Identify Labeling Gaps**	Gap list with clause citations	**Automated**	M3
7	**Gap Addressal**	Content proposed from the approved phrase library; the engineer accepts or edits	**AI-assisted**	M5
8	**Label Content Development**	Label assembled from graph elements, no copy and paste	**AI-assisted**	M5
9	**Redline / Artwork Modification**	Redline generated from the graph diff; layout changes proposed	**AI-assisted**	M5, M4
10	**Stakeholder Review**	Risk-ranked packet: changed elements with clause, source and check results	**Human**	WF
11	**Regulatory / Quality Approval**	Part 11 e-signature; AI never approves	**Human**	WF
12	**Label Revision**	New version committed to the graph and linked to the approval	**Automated**	M0
13	**Change Order Creation**	Drafted with the full impacted list; a person releases it	**Automated**	M1, WF
14	**Document / Configuration Control**	The versioned graph is the configuration record, synced to PLM	**Automated**	M0
15	**Final Quality Check**	CV checks the print proof; QA reviews every flag and signs	**AI-assisted**	M4, WF
16	**Label Release**	E-signed release	**Human**	WF
17	**Manufacturing**	Released files go to the existing print system; CV checks samples	**Automated**	M4
18	**Production Implementation**	Line clearance and first-article sign-off with CV evidence	**Human**	WF

Count: 8 automated, 7 AI-assisted with a human check, 4 human, over the 18 steps plus the new change trigger. Steps 4 and 9 mix automated and AI-assisted work and are counted as AI-assisted. Stakeholder review and the three approvals (11, 16, 18) stay human.

Annexure B: Supporting detail, automation opportunity matrix and impact model

#	Step	Effort	Error risk	Automation	Priority	Module
1	Product / Design Information	2	3	3	8	M0
2	Product Understanding	2	2	2	6	M0
3	Identify Applicable Markets	1	2	2	5	M0
4	Regulatory Requirement Assessment	3	3	2	8	M0, M2
5	Gap Assessment	3	3	3	9	M3
6	Identify Labeling Gaps	2	3	3	8	M1, M3
7	Gap Addressal	2	2	2	6	M5
8	Label Content Development	3	3	2	8	M5
9	Redline / Artwork Modification	3	2	3	8	M5, M4
10	Stakeholder Review	3	2	2	7	WF, M3
11	Regulatory / Quality Approval	2	2	1	5	WF
12	Label Revision	1	2	3	6	M0
13	Change Order Creation	2	2	3	7	M1
14	Document / Configuration Control	2	2	3	7	M0
15	Final Quality Check	3	3	3	9	M4
16	Label Release	1	2	1	4	WF
17	Manufacturing	2	3	2	7	M4
18	Production Implementation	2	2	1	5	WF

Scores 1 (low) to 3 (high) are team estimates from the p.5 pain points, not measurements. Priority is the sum (maximum 9). Steps 5 and 15 score highest.

Impact model: one change, 30 labels × 20 languages (estimate)

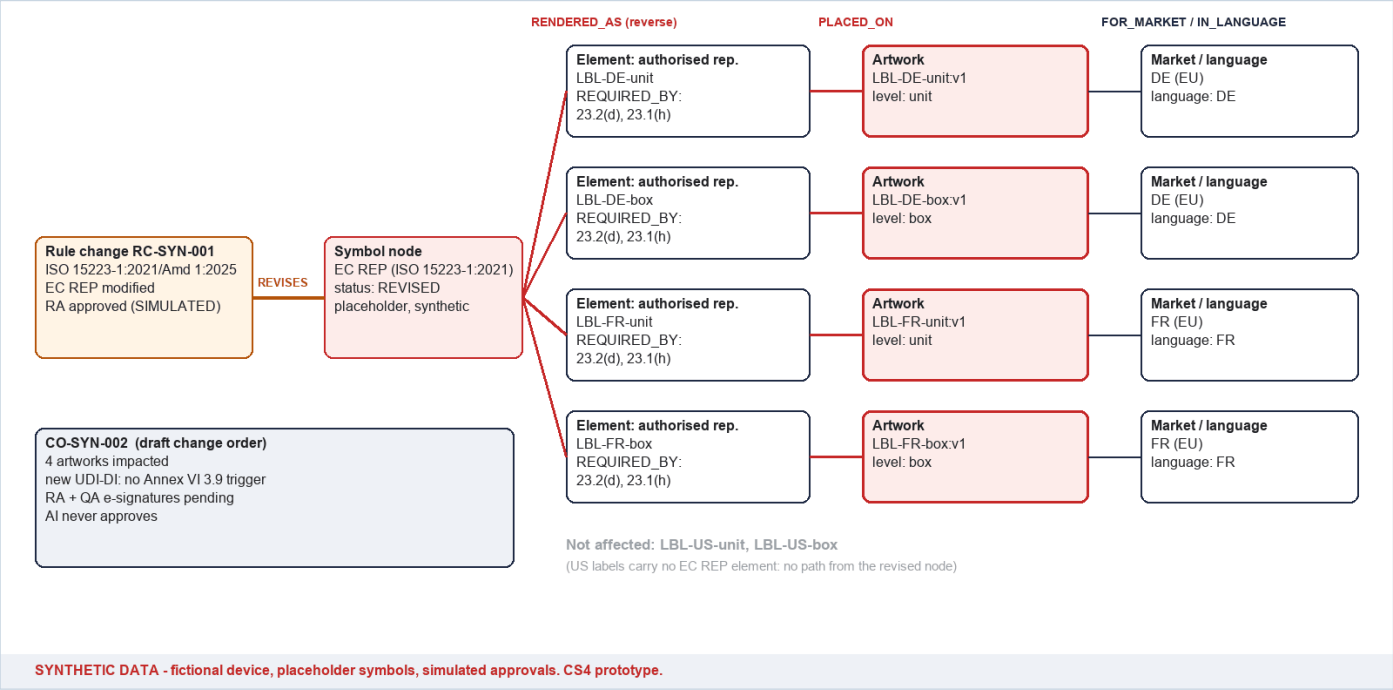
Assumption	Base	Pessimistic
Today: impact assessment	16 h	8 h
Today: redline per label	2 h	1 h
Today: review per language block	5 min × 3 reviewers	3 min × 2 reviewers
Today: final check per label	1 h	0.5 h
ClauseLink: RA confirms impact list	0.5 h	2 h
ClauseLink: layout check per label	0.25 h	0.5 h
ClauseLink: diff-only packet per label	10 min × 2	20 min × 2
ClauseLink: flagged blocks reviewed	10% at 5 min	25% at 5 min
ClauseLink: QA review of CV evidence	0.25 h per label	0.5 h per label

Case	Today	ClauseLink	Saved
Base	256 h	30.5 h	225.5 h (88.1%)
Pessimistic ClauseLink side	256 h	64.5 h	191.5 h (74.8%)
Pessimistic on both sides	113 h	64.5 h	48.5 h (42.9%)

Every label is assumed to carry all 20 languages: 600 language blocks. Effort only. Elapsed time depends on approval queues and is not claimed; error rates and real timings come from a pilot audit trail.

Annexure C: Supporting detail, prototype on synthetic data

Scenario 1: ISO 15223-1 Amd 1 changes the EC REP symbol; one graph query finds the 4 affected labels



Scenario 3: seeded-defect proofs

Proof	Seeded defect	Finding	Clause
(a) US-box	Lot in the human-readable UDI (2410B) differs from the DataMatrix (2410A)	**Printed UDI differs from DataMatrix**	21 CFR 801.40
(b) US-unit	'Rx only' missing on the US label	**Required element missing**	21 CFR 801.109(b)
(c) DE-unit	Old EC REP placeholder (pre-Amd 1) still used	**Symbol edition outdated**	MDR Annex I 23.1(h), 23.2(d)
(d) FR-box	MD symbol (and its caption) missing on an EU label	**Required element missing**	MDR Annex I 23.2(q)

4/4 seeded defects caught. 0 false defect flags across 10 proofs (4 seeded, 6 clean controls). 3 uncertain OCR reads were routed to a human: twice the small "SYN" tag on a placeholder symbol, once an address read just under the 0.80 confidence threshold. Scenario 2 (pack quantity 10 to 5): 3 box labels affected; new UDI-DI flagged for DE and FR under MDR Annex VI 3.9(f); US routed to RA, with a GUDID task under 21 CFR 830.310(b)(14).

Limits of this test

- Symbol templates are drawn by the same code that renders the labels, so matching is easier than on real artwork.
- No rotation, no real ISO artwork, no print-quality grading. Proofs are blurred, noised and JPEG-compressed only.
- All data are synthetic: fictional device, company and GTINs, placeholder symbols, simulated approvals. n = 4 seeded defects is a smoke test, not a validation result.